

AI Agent Engineering Exercise

Design and implement a simple conversational AI agent that can answer questions about SpaceX using the [SpaceX API](#). The focus is not on having a polished UI, but on demonstrating a solid understanding of agentic patterns, prompt design, tool usage, and thoughtful system design.

High-Level Requirements

1. Conversational Agent

- Build an agent that a user can chat with using natural language (e.g., *via a simple console or web interface*).
- The agent should maintain conversational context across multiple turns.

2. Domain: SpaceX

- The agent should be able to discuss SpaceX and answer factual questions about the company based on the live data in the API
- Example queries:
 - “When was the last SpaceX launch?”
 - “What’s the next SpaceX launch and where is it happening?”
 - “How many launches did SpaceX complete in 2024?”
 - “Which rocket was used for the Starlink 9-1 mission?”
 - “Show me all successful Falcon 9 launches.”
 - “What was the outcome of the first Falcon Heavy launch?”
 - “Tell me about the most recent launch from Vandenberg.”

3. Tool Design

- The agent should use one or more tools (functions, APIs, or retrievers) to fetch information from [SpaceX API](#).
- The tools should handle:
 - API request construction
 - Response parsing and data cleaning
 - Error handling and fallbacks
- Example tool calls: (*note, these are not tools you need to incorporate, just examples of tools you could build*)
 - `compare_rockets(rockets: list[str])`
 - `search_missions(query: str)`

4. LLM Integration

- The LLM should interpret user intent, decide when to call a tool, and how to use the result in its response.
- Responses should be clear and grounded in data from the tool output.
- Hardcoding answers or skipping tool usage will not meet the intent of the exercise.

5. Agentic Behavior

- The agent should show some autonomous reasoning — for example:
 - Making multiple tool calls to gather complete information.
 - Asking clarifying questions when user input is ambiguous.
 - Iterating on its own responses to ensure accuracy or completeness.